1 Statements

$\log, h$ base two, ln natural, $h(x) = -x\log x - (1-x)\log(1-x)$; dist Hamming distance on $\{0,1\}^m$; $H(\cdot \mid \cdot)$ conditional Shannon entropy, base two. Except in Sections 4 and 5, where λ and the radius offset vary, fix

$$\lambda = \tfrac{27}{20}, \qquad a = \lambda \ln 2 = 0.93575\ldots, \qquad \tau_r = \tfrac{1}{2} - \tfrac{a}{4r}, \qquad r \geq 2 \text{ integral.}$$

Asymptotics are in $m \to \infty$ (constants may depend on r) or in $r \to \infty$, as indicated. H is *r-degenerate* if every induced subgraph has minimum degree $\leq r$.

Conjecture 1.1 (Erdős–Simonovits [ES84, Blo26])**.** *H bipartite and r-degenerate* $\implies$ $\mathrm{ex}(n,H) \ll_H n^{2-1/r}$.

The bound is attained for $H = K_{r,t}$, $t \geq r$ [KST54]; the best general upper bound is $\mathrm{ex}(n,H) = O(n^{2-1/(4r)})$ [AKS03, Thm. 3.5]. The conjecture holds when one class has maximum degree at most r [Fur91, AKS03], for r-degenerate blow-ups of trees [GJN22], and at $r = 2$ for grids [BJST23] and certain critical 2-degenerate graphs [DGL25]; see [FS13] for the area. It fails at $r = 2$ [OAI26, Ch. 10].

Theorem 1.2. *For every $r \geq 2$ there are a connected bipartite H_r of degeneracy exactly r and $c > 0$ with $\mathrm{ex}(n,H_r) \geq c\, n^{2-\frac{1}{r}+\frac{1}{28r^2}}$ for all large n; at $r = 3$ the exponent is $\frac{5}{3} + \frac{1}{160}$.*

Theorem 1.3. *With A_r, C_r as in* (10),(2) *and λ free:*

(a) at $\tau = \tau_r$, $r^2(C_r - A_r) \to \lambda^4 \ln^3 2/64$ if $\lambda \ln 2 < 1$, and $r^2(C_r - A_r) \to -\infty$ if $\lambda \ln 2 > 1$; the transition is exactly at $2^\lambda = e$;

(b) along $\lambda_r = \frac{1-(\ln r)/r}{\ln 2}$, for each fixed $\theta \in (0,1)$ and $\beta_\theta = A_r + \theta(C_r - A_r)$, $8r^2\varepsilon_r^{\max}(\beta_\theta) \to 1-\theta$ with $\varepsilon_r^{\max}$ as in (4)*; the supremal constant of the method is $1/(8r^2)$, approached as $\beta \to A_r^+$ and not attained.*

Corollary 1.4. *For every $\delta \in (0,1)$ there is $r_0(\delta)$ such that for $r \geq r_0$ there are a connected bipartite $H_{r,\delta}$ of degeneracy exactly r and $c > 0$ with $\mathrm{ex}(n,H_{r,\delta}) \geq c\, n^{2-\frac{1}{r}+\frac{1-\delta}{8r^2}}$ for all large n.*

Verification with Lean 4 and mathlib:

	Lean declaration	File
Thm 1.2	`RDegenerateGraphsTarget.` `rDegenerateExtremalCounterexample_exact`	`proofs/Theorem12.lean`
Thm 1.2, $r = 3$	`ThreeDegenerateGraphsTarget.` `threeDegenerateExtremalCounterexample_sharp`	`proofs/Theorem12r3.lean`
Thm 1.3(a), subcritical	`DegeneracyLaw.` `width_tendsto_unconditional_full`	`proofs/Theorem13a.lean`
Thm 1.3(b)	`DegeneracyLawB.` `eight_rsq_epsMax_theta_tendsto`	`proofs/Theorem13b.lean`
Cor 1.4	`RDegenerateGraphsTarget.` `twoTierExtremalCounterexample`	`proofs/Corollary14.lean`

Each of the five is free of `sorry` and depends only on the three axioms of mathlib's standard foundation, `propext`, `Classical.choice`, `Quot.sound`.

2 Constructions

The forbidden graph. H_r has layers $V_0, \dots, V_{d_r}$ with $|V_0| = N_r$ and $V_{i+1} = \binom{V_i}{r}$, the set of all r-element subsets of V_i, each $S \in V_{i+1}$ (a *child*) joined to each of its r elements (its *parents*); edges join consecutive layers only. Here $d_r = \lceil 100/w_r \rceil + 1$ with $w_r > 0$ from Lemma 4.4, and the root size N_r is chosen large enough in terms of r to meet the two lower bounds imposed in Lemma 3.1 and Proposition 3.3, subject to $N_r \geq 2r^2$; the binding constraint is the first of these, which at $r = 2$ already forces $N_r > 4 \cdot 10^5$, the floor $2r^2$ taking over only from $r \geq 7$. At $r = 2$ this is [OAI26, Ch. 10], cf. [GJN22].

Lemma 2.1. *H_r is connected, bipartite, and has degeneracy exactly r.*

Proof. Layer parity is a proper 2-colouring; every vertex reaches V_0 through parents and any two roots lie in a common child, so H_r is connected. Deleting in the order $V_{d_r}, V_{d_r-1}, \dots, V_0$, a vertex of V_i has as surviving neighbours exactly its $\leq r$ parents, so H_r is r-degenerate. In $H_r[V_0 \cup V_1]$ every child has degree r and every root degree $\binom{N_r-1}{r-1} \geq r$, so the minimum degree is $\geq r$ and H_r is not $(r-1)$-degenerate.[1] □

The host. For $\tau \in (0, \frac{1}{2})$, $\beta \in (0,1)$ let $G = G(m, \tau, \beta)$ be obtained from two disjoint copies L, R of $\{0,1\}^m$ by joining $x \in L, y \in R$ whenever $\operatorname{dist}(x,y) \leq \tau m$ and retaining each vertex independently with probability $2^{-\beta m}$. Write n for the number of retained vertices and E for the number of edges with both ends retained.

Lemma 2.2. *If $1 + h(\tau) - 2\beta > 0$ then with probability $1 - o(1)$*

$$n = 2^{(1-\beta)m + O(\log m)}, \qquad E = 2^{(1+h(\tau)-2\beta)m + O(\log m)}; \tag{1}$$

hence $E \geq n^{\gamma - o(1)}$ with $\gamma = \frac{1+h(\tau)-2\beta}{1-\beta}$, and $\gamma > 2 - \frac{1}{r}$ iff

$$\beta < C_r(\tau) := r\,h(\tau) - (r-1). \tag{2}$$

Proof. $n \sim$ Bin with mean $\mu_n = 2^{(1-\beta)m+1}$ and $\operatorname{Var}(n) \leq \mu_n$, so Chebyshev gives $n = (1 + o(1))\mu_n$. The number of eligible pairs is $T = 2^m \sum_{k \leq \tau m} \binom{m}{k}$, and $\binom{m}{\lfloor \tau m \rfloor} \leq \sum_{k \leq \tau m} \binom{m}{k} \leq (\tau m + 1)\binom{m}{\lfloor \tau m \rfloor}$ for $\tau < \frac{1}{2}$, so $T = 2^{(1+h(\tau))m + O(\log m)}$ and $\mu_E = T 2^{-2\beta m} \to \infty$. Two edge slots are dependent only if they share a vertex, and the number of ordered such pairs is at most $2T \sum_{k \leq \tau m} \binom{m}{k} = 2^{(1+2h(\tau))m + O(\log m)}$, each surviving with probability $2^{-3\beta m}$; hence

$$\operatorname{Var}(E) \leq \mu_E + 2^{(1+2h(\tau)-3\beta)m + O(\log m)} = \mu_E + \mu_E^2\, 2^{-(1-\beta)m + O(\log m)} = o(\mu_E^2), \tag{3}$$

and Chebyshev applies again. On the intersection of the two events $\log E / \log n = \gamma - o(1)$. Finally $\gamma > 2 - \frac{1}{r}$ rearranges, using $1 - \beta > 0$, to $r(1 + h(\tau) - 2\beta) > (2r-1)(1-\beta)$, i.e. $\beta < r\,h(\tau) - (r-1)$.[2][3] □

For $\beta < C_r(\tau)$ set the *excess*

$$\varepsilon_r^{\max}(\beta) := \gamma - \left(2 - \tfrac{1}{r}\right) = \frac{C_r(\tau) - \beta}{r(1-\beta)}, \tag{4}$$

decreasing in β on the window, since $C_r < 1$ for $\tau < \frac{1}{2}$. The construction realizes one n per m, with consecutive values in ratio $2^{1-\beta+o(1)}$; padding with isolated vertices extends any bound $\operatorname{ex}(n, H_r) \geq n^{2-1/r+\varepsilon}$ from that sequence to all large n at the cost of c alone. The $o(1)$ losses in Lemma 2.2 and (4) are absorbed by the slack factor $1 - \eta$, $\eta = \frac{1}{100}$, of Section 6.

[1] `not_isDegenerate_of_layer_witness`, `proofs/lib/ExactDegeneracy.lean`.
[2] `sampledREdgeEntropyRate_gt`, `proofs/lib/SamplingR.lean`.
[3] `exists_good_retention_edge_lower`, `proofs/lib/SamplingR.lean`.

3 The entropy ceiling

Fix an embedding of H_r in the host: an assignment of strings in $\{0,1\}^m$ to $V(H_r)$ with consecutive layers on opposite sides and every child within τm of each of its parents. All quantities are empirical, computed from the string arrays of two consecutive layers V_i, V_{i+1}; put $L = |V_i|$, $M = |V_{i+1}| = \binom{L}{r}$. Let q_j be the fraction of ones among the parents' j-th bits, v_j the same for the children, and

$$\Phi(V_i) := \frac{1}{m}\sum_{j=1}^{m} h(q_j) \in [0,1]. \tag{5}$$

With Z the string of a uniform child and $X_1, \dots, X_r$ those of its parents in uniform random order, put

$$\mathcal{H}(V_{i+1} \mid V_i) := \frac{1}{m}\sum_{j=1}^{m} H\big(Z^{(j)} \mid X_1^{(j)}, \dots, X_r^{(j)}\big), \qquad D(V_i, V_{i+1}) := \frac{1}{rm}\,\mathbb{E}\sum_{s=1}^{r} \operatorname{dist}(Z, X_s), \tag{6}$$

the conditioning being on the parents' j-th bits alone, so that $\mathcal{H}$ is the coordinatewise quantity counted by the type profiles of Lemma 3.1; by the chain rule it dominates $\frac{1}{m}H(Z \mid X_1, \dots, X_r)$, as in (11). Proximity gives $D \le \tau$.

Lemma 3.1. *Fix $\delta > 0$. Call an embedding* low *if $\mathcal{H}(V_{i+1} \mid V_i) \le \beta - \delta$ for some i. Then with probability $1 - o(1)$ no low embedding is entirely retained.*

Proof. Fix i and consider triples (parent array, type profile, child array), a *profile* recording for each $j \in [m]$ and each parent popcount $k \in \{0, \dots, r\}$ the number of children with k one-parents at j whose own j-th bit is one. There are at most 2^{Lm} parent arrays and $(M+1)^{(r+1)m}$ profiles, and given both at most $2^{(\beta-\delta)mM}$ child arrays with $\mathcal{H}(V_{i+1} \mid V_i) \le \beta - \delta$: coordinatewise the count is a product of binomial coefficients whose logarithm is exactly mM times the empirical conditional entropy of the profile.[4] The M children are distinct vertices of one side, so all are retained with probability $2^{-\beta mM}$. Hence the expected number of surviving low layer pairs is at most

$$\sum_{i<d_r} 2^{Lm}(M+1)^{(r+1)m} 2^{(\beta-\delta)mM} 2^{-\beta mM} = \sum_{i<d_r} 2^{m(L+(r+1)\log(M+1)-\delta M)} \le 2d_r 2^{-m}, \tag{7}$$

provided $L + (r+1)\log(M+1) - \delta M < -1$ for each layer, which holds once N_r is large in terms of r, δ, since $M = \binom{L}{r}$ is polynomial of degree $r \ge 2$ in L against $O(L)$ elsewhere and layer sizes increase. Markov concludes.[5][6][7] □

For $q, v \in [0,1]$ set

$$G_r(q,v) = \frac{h(q)}{2} + \sum_{k=0}^{r} \binom{r}{k} q^k (1-q)^{r-k} \log\Big(\sqrt{1-v}\,2^{-\lambda k/r} + \sqrt{v}\,2^{-\lambda(r-k)/r}\Big). \tag{8}$$

Lemma 3.2. *For consecutive embedded layers with $L \ge 2r^2$,*

$$\mathcal{H}(V_{i+1} \mid V_i) \le \lambda D(V_i, V_{i+1}) + \sup_{[0,1]^2} G_r + \frac{\Phi(V_{i+1}) - \Phi(V_i)}{2} + \kappa_r(L), \tag{9}$$

[4] `rChildArraysOfRealizedProfile_card_le`, `proofs/lib/ProfilesR.lean`.
[5] `badRChildRetentionEvent`, `proofs/lib/SamplingR.lean`.
[6] `badRLayerRetentionEvent`, `proofs/lib/SamplingR.lean`.
[7] `exists_good_retention`, `proofs/lib/SamplingR.lean`; it carries the event of Lemma 2.2 in the same conjunction.

with $\kappa_r(L) \to 0$ as $L \to \infty$ independently of m. In particular, with

$$A_r(\lambda) := \lambda\tau + \sup_{[0,1]^2} G_r, \tag{10}$$

one has $\mathcal{H}(V_{i+1} \mid V_i) \le A_r + \frac{1}{2}(\Phi(V_{i+1}) - \Phi(V_i)) + \kappa_r(L)$.

Proof. The chain rule gives

$$\begin{aligned} \frac{1}{m} H\big(Z \mid X_1, \dots, X_r\big) &= \frac{1}{m} \sum_{j=1}^{m} H\big(Z^{(j)} \mid X_1, \dots, X_r, Z^{(<j)}\big) \\ &\le \frac{1}{m} \sum_{j=1}^{m} H\big(Z^{(j)} \mid X_1, \dots, X_r\big) \\ &\le \frac{1}{m} \sum_{j=1}^{m} H\big(Z^{(j)} \mid X_1^{(j)}, \dots, X_r^{(j)}\big) \;=\; \mathcal{H}(V_{i+1} \mid V_i), \end{aligned} \tag{11}$$

the first inequality dropping the earlier child bits and the second coarsening each parent string to its j-th bit, of which it is a function; both only increase entropy. Both bounds of this section constrain $\mathcal{H}$.

Fix j and write $q = q_j$, $v = v_j$. Let $P(k)$ be the law of the parent popcount at j and p_k the conditional probability that the child bit is one given popcount k, so that $\sum_k P(k)p_k = v$. For each k, Gibbs' inequality $\sum_b p_b \log(w_b/p_b) \le \log \sum_b w_b$ applied to the conditional law $(1-p_k, p_k)$ with the weights $w_0 = \sqrt{1-v}\, 2^{-\lambda k/r}$, $w_1 = \sqrt{v}\, 2^{-\lambda(r-k)/r}$ carrying the square root of the *marginal* gives

$$h(p_k) + (1-p_k)\log\sqrt{1-v} + p_k \log\sqrt{v} - \frac{\lambda}{r}\,\mathbb{E}_k[d] \;\le\; \log\Big(\sqrt{1-v}\, 2^{-\lambda k/r} + \sqrt{v}\, 2^{-\lambda(r-k)/r}\Big), \tag{12}$$

d denoting the number of the r parent bits disagreeing with the child bit. Average (12) against P. Since $\sum_k P(k)p_k = v$ and $\sum_k P(k) = 1$, the two square-root terms collapse:

$$\sum_k P(k)\Big[(1-p_k)\log\sqrt{1-v} + p_k \log\sqrt{v}\Big] = (1-v)\log\sqrt{1-v} + v\log\sqrt{v} = -\tfrac{1}{2}h(v). \tag{13}$$

Averaging over j, these terms contribute $-\frac{1}{2}\Phi(V_{i+1})$ and the exponent terms contribute λD. Averaging over r-subsets makes k hypergeometric; replacing it by $\mathrm{Bin}(r, q_j)$ costs the without-replacement coupling error $\kappa_r(L)$ under $L \ge 2r^2$.[8] Adding and subtracting $\frac{1}{2}h(q_j)$, the added term and the binomial average of (12) are exactly $G_r(q_j, v_j)$, so $\frac{1}{m}\sum_j G_r(q_j, v_j) \le \sup G_r$, while the subtracted terms average to $-\frac{1}{2}\Phi(V_i)$.[9][10][11] □

Proposition 3.3. *Let $\beta > A_r(\lambda)$ and $d_r \ge \lceil 100/w_r \rceil + 1$ with $0 < w_r \le 100(\beta - A_r)$. Then with probability $1 - o(1)$ the retained host contains no copy of H_r.*

Proof. Put $\delta = \frac{1}{4}(\beta - A_r)$ and let N_r be large enough that $\kappa_r(L) \le \delta$ for $L \ge N_r$. Suppose a copy of H_r is entirely retained, with layers $V_0, \dots, V_{d_r}$. By Lemma 3.1 we may assume $\mathcal{H}(V_{i+1} \mid V_i) > \beta - \delta$ for all i, whence by Lemma 3.2 (applicable since $|V_i| \ge N_r \ge 2r^2$ and layer sizes increase)

$$\Phi(V_{i+1}) - \Phi(V_i) > 2(\beta - A_r - 2\delta) = \beta - A_r \ge \frac{w_r}{100} \qquad (0 \le i < d_r). \tag{14}$$

[8] `worRExpectation_error`, `proofs/lib/BridgeR.lean`.
[9] `rCoordinateConditionalEntropy_empirical_bound`, `proofs/lib/BridgeR.lean`.
[10] `rChildArrayEntropy_empirical_bound`, `proofs/lib/BridgeR.lean`.
[11] `RGenericKernel.TypeEntropyBound`, `proofs/lib/KernelR.lean`.

But $\Phi \in [0,1]$, while these d_r increments raise it by more than $(\frac{100}{w_r}+1)\frac{w_r}{100} > 1$; contradiction. [12][13] □

Thus any β with $A_r(\lambda) < \beta < C_r(\tau)$ yields, for all large m, a host that is H_r-free and has $E \geq n^{2-\frac{1}{r}+\varepsilon-o(1)}$ edges, $\varepsilon = \varepsilon_r^{\max}(\beta)$: both events hold with probability $1-o(1)$, so some outcome realizes both.

4 Nonemptiness of the window

Both thresholds are $1-\Theta(1/r)$, so the comparison is in the lower-order terms.

Lemma 4.1. *For $a > 0$ and $F_a(q) = h(q) + \log(e^{-2aq} + e^{-2a(1-q)})$ on $(0,1)$, $F_a''(q)\ln 2 \leq 4(a^2-1)$; in particular F_a is strictly concave when $a < 1$, i.e. when $2^\lambda < e$.*

Proof. $h''(q)\ln 2 = -\frac{1}{q(1-q)} \leq -4$. Factoring out e^{-a}, $g_a(q) := \log(e^{-2aq}+e^{-2a(1-q)}) = 1 - a\log e + \log\cosh(a(1-2q))$, and with $u = a(1-2q)$, $\frac{d^2}{dq^2}\ln\cosh u = 4a^2\operatorname{sech}^2 u \leq 4a^2$; add.[14] □

Lemma 4.2. *For $\lambda \leq \frac{27}{20}$ and every $r \geq 2$,*

$$\sup_{[0,1]^2} G_r = G_r\big(\tfrac12,\tfrac12\big) = 1 - \frac{\lambda}{2} + \mathbb{E}\log\cosh\Big(\frac{aS_r}{2r}\Big), \qquad S_r = 2\,\mathrm{Bin}\big(r,\tfrac12\big) - r, \tag{15}$$

so $A_r = \lambda\tau + 1 - \frac{\lambda}{2} + \mathbb{E}\log\cosh\big(\frac{aS_r}{2r}\big)$.

Proof. At $v = \frac12$ the k-th summand of (8) is $\log(2^{-1/2}(2^{-\lambda k/r} + 2^{-\lambda(r-k)/r})) = \frac12 - \frac{\lambda}{2} + \log\cosh(\frac{a}{2r}(r-2k))$; at $q = \frac12$ the weights make $r - 2k \sim S_r$, and $h(\frac12)/2 = \frac12$, giving the stated value. For the maximization substitute $v = \sin^2\theta$, $\theta \in [0, \frac{\pi}{2}]$, so that $G_r(q,v) = \frac12 h(q) + B_r[\psi_a(\cdot,\theta)](q)$ with B_r the Bernstein operator and $\psi_a(u,\theta) = \log(\cos\theta\, 2^{-\lambda u} + \sin\theta\, 2^{-\lambda(1-u)})$; note $G_r(q,v) = G_r(1-q,1-v)$. The master inequality[15] states that for $a < 1$, with $t_s = \arctan e^{a(2q-1)}$, which satisfies $|t_s - \frac{\pi}{4}| \leq a|q-\frac12|$,

$$G_r(q,v) \leq 1 - \frac{\lambda}{2} - \frac{(1-a^2)(q-\frac12)^2 + \frac12(\theta - t_s)^2}{\ln 2} + \frac{a^2}{8r\ln 2}\,: \tag{16}$$

the margin in q is the concavity margin $4(1-a^2)$ of Lemma 4.1; the margin in θ comes from splitting ψ_a into a radial part maximized at $\theta = t_s(q)$ and an angular part concave in θ; and $B_r[f](q) - f(q) \leq \frac12\|f''\|_\infty \operatorname{Var}(k/r) \leq \frac{a^2}{8r\ln 2}$ is the Bernstein excess. Since the centre exceeds $1-\frac{\lambda}{2}$ by $\mathbb{E}\log\cosh(\frac{aS_r}{2r}) = \frac{a^2}{8r\ln 2}(1 - O(\frac1r))$ (Lemma 4.4), which matches that excess to first order, (16) settles the comparison except where both quadratic margins are $O(1/r^2)$. Split along $|\theta - \frac{\pi}{4}| \leq \rho$, $\rho = 0.99\cdot\frac12\arccos\kappa$, $\kappa = \sqrt{a^2/(2-a^2)}$: off the strip, $|t_s - \frac{\pi}{4}| \leq a|q-\frac12|$ turns the angular deficit into a fixed margin dominating $\frac{a^2}{8r\ln 2}$; inside, a sharpened second-derivative estimate along the rotated direction shows G_r concave with the centre its only critical point. At $\lambda \leq \frac{27}{20}$ both branches hold for every $r \geq 2$.[16] □

Remark 4.3. A quantified form, valid for each fixed λ with $\lambda\ln 2 < 1$ and all large r [17] is what Theorem 1.3 uses: the schedule λ_r exceeds $\frac{27}{20}$ once $r \geq 65$.

[12] `potential_layers_impossible`, `proofs/lib/BridgeR.lean`.
[13] Depth bookkeeping, `proofs/lib/LedgerR.lean`.
[14] `DegeneracyLaw.Ffun_second_deriv`, `proofs/Lemma41.lean`, over `proofs/lib/EntropyConcavity.lean`.
[15] `Gfun_le_center_margin`, `proofs/lib/BernsteinStrip.lean`.
[16] `DegeneracyLaw.Lemma42.supG_eq_center`, `proofs/Lemma42.lean`.
[17] `supG_eq_center_quant`, `proofs/lib/CenterMaxThreshold.lean`.

Lemma 4.4. *At $\tau = \frac{1}{2} - \frac{c}{r}$, $c > 0$ fixed,*

$$C_r(\tau) - A_r = -\frac{(4c-a)^2}{8\ln 2} \cdot \frac{1}{r} + O\Big(\frac{1}{r^2}\Big) \qquad (r \to \infty), \tag{17}$$

so the tuned radius $c = a/4$ is forced, and there

$$w_r := C_r(\tau_r) - A_r = \frac{\lambda^4 \ln^3 2}{64 r^2}\Big(1 + O\big(\tfrac{1}{r}\big)\Big) > 0 \tag{18}$$

for all large r. At $\lambda = \frac{27}{20}$ one has $w_r \geq 0.00603/r^2$ for every *$r \geq 2$, and $w_3 \geq 0.0098/9$.*

Proof. With $x = c/r$ and $\ln 2 - H_e(\frac{1}{2} - x) = \sum_{k\geq 1} \frac{(2x)^{2k}}{2k(2k-1)} = 2x^2 + \frac{4}{3}x^4 + \cdots$,

$$C_r(\tau) = r\, h\big(\tfrac{1}{2} - \tfrac{c}{r}\big) - (r-1) = 1 - \frac{2c^2}{\ln 2} \cdot \frac{1}{r} - \frac{4c^4}{3\ln 2} \cdot \frac{1}{r^3} - \cdots . \tag{19}$$

By Lemma 4.2, $A_r = 1 - \frac{\lambda c}{r} + \mathbb{E}\log\cosh(tS_r)$ with $t = \frac{a}{2r}$, since $\lambda\tau + 1 - \frac{\lambda}{2} = 1 - \frac{\lambda c}{r}$. The sandwich $\frac{t^2}{2} - \frac{t^4}{12} \leq \ln\cosh t \leq \frac{t^2}{2} - \frac{t^4}{12} + \frac{t^6}{45}$ with $\mathbb{E}S_r^2 = r$, $\mathbb{E}S_r^4 = 3r^2 - 2r$, $\mathbb{E}S_r^6 = O(r^3)$ gives

$$\mathbb{E}\log\cosh(tS_r) = \frac{1}{\ln 2}\Big(\frac{a^2}{8r} - \frac{a^4}{64r^2} + O\big(\tfrac{1}{r^3}\big)\Big), \tag{20}$$

whence, using $\lambda = a/\ln 2$,

$$C_r - A_r = \frac{1}{\ln 2}\Big(-2c^2 + ac - \frac{a^2}{8}\Big)\frac{1}{r} + \frac{a^4}{64\ln 2} \cdot \frac{1}{r^2} + O\Big(\frac{1}{r^3}\Big), \tag{21}$$

and $-2c^2 + ac - \frac{a^2}{8} = -\frac{(4c-a)^2}{8}$. At $c = a/4$ the $1/r$ term vanishes and $\frac{a^4}{64\ln 2} = \frac{\lambda^4 \ln^3 2}{64}$: the quadratic parts of the two thresholds cancel exactly and the width is positive because the quartic term of $\log\cosh$ is negative. The uniform bounds at $\lambda = \frac{27}{20}$ are certified in exact rational arithmetic over the same sandwich.[18][19] □

5 Proof of Theorem 1.3 and Corollary 1.4

Proof of Theorem 1.3. (a) For $\lambda\ln 2 < 1$ the $1/r$ terms cancel at τ_r (Lemma 4.4, which uses $\lambda\ln 2 < 1$ only through Remark 4.3) and the $\log\cosh$ sandwich encloses $r^2(C_r - A_r)$ between two sequences with limit $\lambda^4\ln^3 2/64$. For $\lambda\ln 2 > 1$ concavity of F_a fails on a central interval — Lemma 4.1 is sharp, $F_a''\ln 2 \to 4(a^2-1)$ as $q \to \frac{1}{2}$ — so $\sup G_r$ exceeds its centre value by a positive constant depending only on a, and $r^2(C_r - A_r) \to -\infty$;[20] the transition is at $2^\lambda = e$.[21]

(b) By (4), $\varepsilon_r^{\max}(\beta_\theta) = \frac{(1-\theta)(C_r - A_r)}{r(1-\beta_\theta)}$. Along $\lambda_r = \frac{1-(\ln r)/r}{\ln 2}$, subcritical for every r and critical in the limit, part (a) gives $r^2(C_r - A_r) \to \lambda^4\ln^3 2/64 \to \frac{1}{64\ln 2}$, while Lemma 4.2 gives $1 - A_r = \frac{\lambda a}{4r} - \mathbb{E}\log\cosh(\frac{aS_r}{2r}) = \frac{\lambda a}{8r}(1 + o(1))$, so $r(1-\beta_\theta) \to \frac{\lambda^2\ln 2}{8} \to \frac{1}{8\ln 2}$. Hence $8r^2\varepsilon_r^{\max}(\beta_\theta) \to 8(1-\theta)\frac{1/(64\ln 2)}{1/(8\ln 2)} = 1 - \theta$. As $\varepsilon_r^{\max}$ decreases in β, the constant $1/(8r^2)$ is approached as $\theta \to 0$ and not attained. □

[18] `proofs/lib/WindowLowerBound.lean` and `proofs/lib/WindowUpperBound.lean`.
[19] `DegeneracyLaw.width_pos_all`, `proofs/lib/AssemblyR.lean`.
[20] `DegeneracyLawSuper.rsq_width_tendsto_atBot`, `proofs/lib/SuperCritical.lean`.
[21] `proofs/lib/WindowLimit.lean`.

Proof of Corollary 1.4. Fix $\delta \in (0,1)$, put $\theta = \eta = \delta/4$, and run Sections 2–3 at $(\lambda_r, \tau_r, \beta_\theta)$ with depth $d_{r,\delta} = \lceil 2/(\theta w_r) \rceil + 1$, $w_r = C_r - A_r$ at λ_r; $H_{r,\delta}$ is the layered graph with that depth.[22] Every step is stated for arbitrary subcritical Gibbs weight and arbitrary β in the window, and Proposition 3.3 applies with increments exceeding $2(\beta_\theta - A_r - 2 \cdot \frac{1}{4}\theta w_r) = \theta w_r$. The quantified Lemma 4.2 applies along the schedule since its threshold obeys $R(\lambda_r) = O(r/\ln r) = o(r)$ (the strip half-width scales as $\sqrt{1-a}$ and $1 - a_r = (\ln r)/r$),[23] and the window is nonempty for large r by Theorem 1.3(a).[24] By Theorem 1.3(b),[25] $8r^2 \varepsilon_r^{\max}(\beta_\theta) \to 1 - \frac{\delta}{4}$; enlarging $r_0(\delta)$ so that it exceeds $1 - \frac{\delta}{2}$ and carrying the slack $1 - \eta$,

$$\big(1 - \tfrac{\delta}{4}\big)\varepsilon_r^{\max}(\beta_\theta) > \big(1 - \tfrac{\delta}{4}\big)\big(1 - \tfrac{\delta}{2}\big)\frac{1}{8r^2} = \Big(1 - \tfrac{3\delta}{4} + \tfrac{\delta^2}{8}\Big)\frac{1}{8r^2} > \frac{1-\delta}{8r^2}. \tag{22}$$

Padding extends the bound to all large n.[26] □

6 Proof of Theorem 1.2

Proof. Fix $r \geq 2$, $\lambda = \frac{27}{20}$, $\tau = \tau_r$, and $\beta = A_r + \frac{w_r}{100}$, just inside the exclusion edge, which is the optimal end of the window by Theorem 1.3(b). By Lemma 4.4, $w_r > 0$, so $A_r < \beta < C_r$; by Lemma 2.2 the retained host has at least $n^{2-\frac{1}{r}+\varepsilon}$ edges for any $\varepsilon < \varepsilon_r^{\max}(\beta)$ with probability $1 - o(1)$, and by Proposition 3.3 it is H_r-free with probability $1 - o(1)$, so some host is both. Carrying $1 - \eta$ at $\eta = \frac{1}{100}$ and using $C_r - \beta = (1 - \frac{1}{100})w_r$,

$$\varepsilon = \Big(1 - \tfrac{1}{100}\Big)\varepsilon_r^{\max}(\beta) = \Big(1 - \tfrac{1}{100}\Big)^2 \frac{w_r}{r(1-\beta)} \geq \frac{0.99^2 \times 0.00603}{0.1637\, r^2} \geq \frac{1}{28r^2}, \tag{23}$$

by the certified bounds $w_r \geq 0.00603/r^2$ (Lemma 4.4) and $r(1-\beta) \leq 0.1637$, uniform in r, both in exact rational arithmetic;[27][28] the middle bound is $0.0361\ldots/r^2$, so rounding to $\frac{1}{28r^2} = 0.0357\ldots/r^2$ loses about one percent. Padding extends the bound to all large n. At $r = 3$ the same chain with $w_3 \geq 0.0098/9$ gives $\varepsilon_3 \geq \frac{1}{160}$.[29] Lemma 2.1 supplies the properties of H_r. □

[22] The host pipeline re-instantiated at $(\lambda_r, \theta{=}\eta{=}\frac{\delta}{4})$, `proofs/lib/AssemblySchedHost.lean`.
[23] `R_of_lamR_le`, `proofs/lib/LedgerTuned.lean`.
[24] `DegeneracyLawB.width_tuned_tendsto`, `proofs/lib/LedgerTuned.lean`.
[25] `eight_rsq_epsMax_theta_tendsto`, `proofs/Theorem13b.lean`.
[26] The schedule parameters, with their eventual admissibility and the certified bound, are `schedParams_admissible`, `proofs/lib/AssemblySched.lean`.
[27] `DegeneracyLaw.Lemma44.width_ge`, `proofs/Lemma44.lean`, in the uniform form of `proofs/lib/AssemblyR.lean`.
[28] $r(1-\beta) \leq 0.1637$, `proofs/lib/LedgerSharp.lean`.
[29] `proofs/lib/AssemblyR.lean`.

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